

# Nine result families

Four evidence questions · read every card literally.



A · analytic



F · formal subclaim



B · bounded executable



S · report / integration state

## FOUNDATIONAL AND FINITE-LAW RESULTS

1

### Foundational categorical Sx audit

A exact semantics + counterexamples

F Lean factorization slice

B · witnesses + bounded checks

S · SCOPED REPORT

2

### Finite-alphabet plug-in convergence

A exact-real convergence

F Lean continuity core

B · fixtures + Rust comparison

S · SCOPED REPORT

3

### Support-change-tolerant averaged Sx continuity

A exact continuity moduli

F Lean algebraic subclaims

B · 18 law pairs + replay

S · SCOPED REPORT

4

### Dependency-color concentration and local bounds

A probability theorem

F deterministic subclaims

B · oracle + Rust checks

S · SCOPED REPORT

5

### Exact two-source Sx assurance

A count-to-atom + product theorem

F Lean bridge + algebra

B · certifier + bounded routes

S · INTEGRATION OPEN

## HIGHER-SOURCE, NUMERICAL, AND CONTINUOUS RESULTS

6

### SxPID3 source-marginal factorization and bounded exact audit

A source-only factor

F Lean generic theorem

B · 20,348 tables · two routes

S · INTEGRATED / SCOPED

7

### Represented-binary64 and quantizer assurance

A exact represented sum

F no end-to-end proof

B · oracles + bounded tests

S · ASSURANCE REPORT

8

### KSG integer-harmonic local arithmetic

A harmonic identity + sharp bound

F Lean + Z3 subclaims

B · 8,198-row corpus

S · CORE GO / INTEGRATION NO-GO

9

### Common-radius manifold small-ball bridge

A conditional limit + boundary test

F no machine proof

B · no finite-sample evidence

S · ESTIMATOR OPEN

### NO PROOF TRANSFER BY PROXIMITY

A theorem, formal subclaim, bounded corpus, and publication state answer different questions.